

Lab Vision Systems (MA-INF 4308)

Lab Project: Weakly Supervised ML for Rapeseed Feeding-Damage Scoring

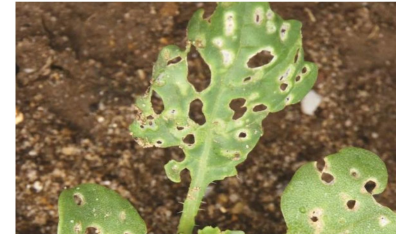
15.07.2026

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Problem Setting

- **Cabbage stem flea beetle (CSFB)**
 - > destroys young rapeseed plants
 - > shot-holing and pitting
- **Goal:** Find plant types that are **resistant** to CSFB
 - > damage quantification
 - > feature computation & analysis



Typical leaf feeding damage on young seedling of oilseed rape in the field

Plant Age & Development

- Classified according to **BBCH** scale



BBCH 10 ⇔ 2 **cotyledons** developed

BBCH 11 ⇔ 1 **true leaf** developed

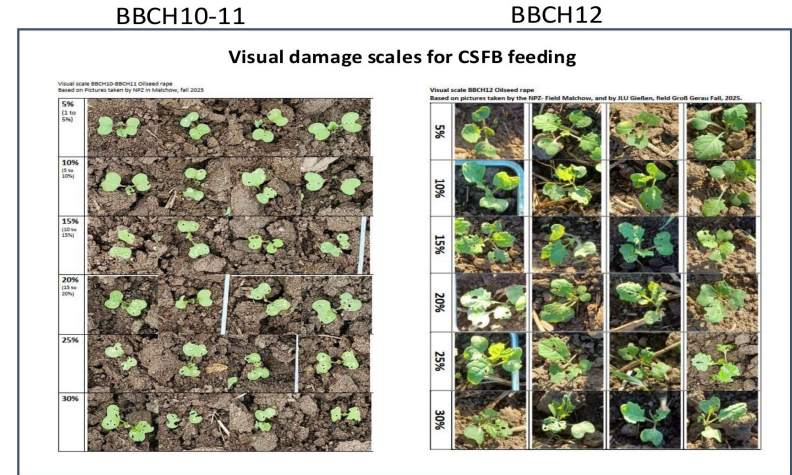
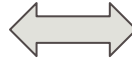
BBCH 12 ⇔ 2 **true leaves** developed

(...)



Damage Quantification

- Given an image, quantify average damage in %



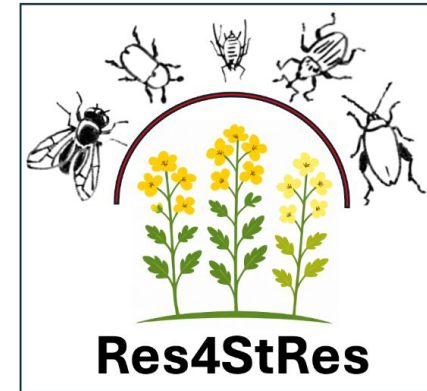
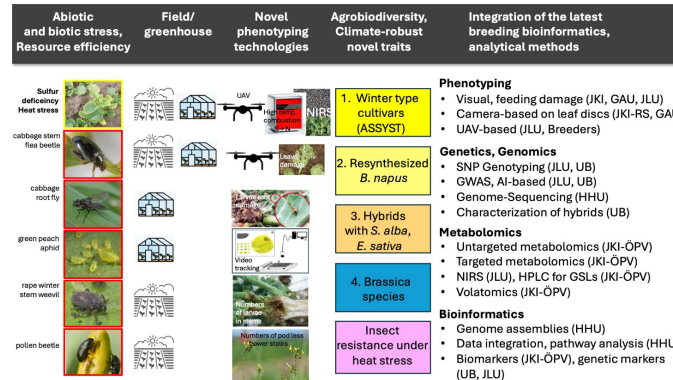
Feature Computation

- Additionally, we are interested in computing features for further analysis:
 - > number of plants in image (and inside metal frame)
 - > average number/area of shot holes per plant
 - > average number/area of yellow/brown spots
 - > number of cotyledons and true leaves per plant

Project Info

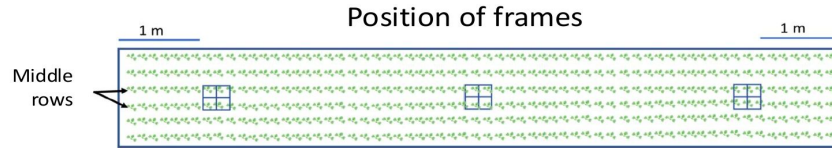
- Part of the **Res4StRes** project

“Novel resources for resistance breeding against insect pests and heat stress under low cropping intensities in oilseed rape”



Provided Data

- digital pictures from plots (1,5 x 11 m) at 6 different locations



**Three pictures per plot,
300 or 400 plots per location
at one or two time-points**

Partner	Location	Experiment	Field plot layout	Sowing date	1st CSFB Phenotyping	2nd CSFB Phenotyping
JLU	Groß-Gerau	Sulfur	4 x 100	8.9.2025	-	14.10. BBCH15 (1271)
JLU	Groß-Gerau	Insects	6 x 50	1.10.2025	21.10. BBCH10 (937)	10.11. BBCH13 (937)
NPZI	Malchow	Insects	10 x 30	29.8.2025	12.9. BBCH10 (900)	19.9. BBCH13 (900)
DSV	Asendorf	Insects	5 x 60	3.9.2025	15.-17.9. BBCH11 (910)	30.0./1.10. BBCH15 (900)
JLU	Weilburger Grenze	Insects	5 x 60	18.9.2025	7.10. BBCH10 (900)	20./24.10. BBCH15 (360+541)
JLU	Rauischholzhausen	Insects	3 x 100	2.9.2026	-	29.9. BBCH13 (900)



Provided Data

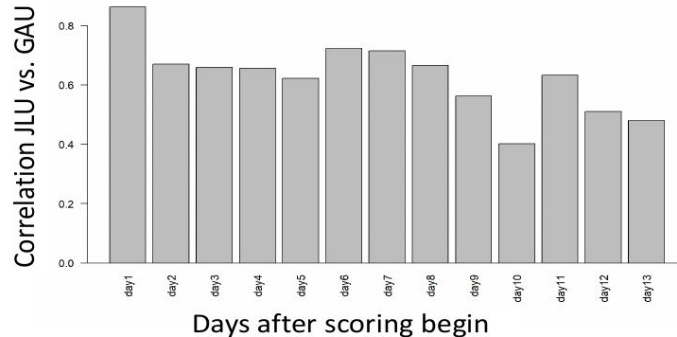
- Total pictures:
8946
- Labelled with scores:
2747
- Labelled independently
by 2 institutions:
937
- With “high quality” labels:
470/937

Partner	Location	1st CSFB Phenotyping	2nd CSFB Phenotyping	Zip folder of picture files, first part of file name identifies sampling date and BBCH in columns 3 and 4 (number of contained jpg files)	Folder size	Visual scoring finished, files (in main folder) and in folder RFSB-Phenotyping_training_set.zip
JLU	Groß-Gerau	-	14.10. BBCH15	2025_10_14_RSFB-Phenotyping_GG_Sulfur_JLU.zip (1271)	9.42 GB	-
JLU	Groß-Gerau	21.10. BBCH10	10.11. BBCH13	2025_10_21_RSFB-Phenotyping_GG1_JLU.zip (937) 2025_11_10_RSFB-Phenotyping_GG2_JLU.zip (937)	7.99 GB 7.52 GB	2025_10_21_RSFB-Phenotyping_GG1_scores.csv (2 x 937) -
NPZI	Malchow	12.9. BBCH10	19.9. BBCH13	2025_09_12_RSFB_01_NPZI.zip (900) 2025_09_19_RSFB_02_NPZI.zip (900)	5.35 GB 4.48 GB	- -
DSV	Asendorf	15.-17.9. BBCH11	30.9./1.10. BBCH15	2025_09_15_Res4StRes_T1_DSV.zip (910) 2025_09_30_Res4StRes_T2_DSV.zip (900)	7.10 GB 7.54 GB	2025_09_15_Res4StRes_T1_DSV_scores.csv (910) -
JLU	Weilburger Grenze	7.10. BBCH10	20./24.10. BBCH15 BBCH15	2025_10_07_RSFB-Phenotyping_WG1_JLU.zip (900) 2025_10_20_RSFB-Phenotyping_WG2_FirstHalf_JLU.zip (360) 2025_10_24_RSFB-Phenotyping_WG2_SecondHalf_JLU.zip (541)	6.72 GB 2.85 GB 4.21 GB	2025_10_07_RSFB-Phenotyping_WG1_JLU_scores.csv (900) - -
JLU	Rauischholzhausen	-	29.9. BBCH13	2025_09_29_RSFB-Phenotyping_RHH1_JLU.zip (900)	5.53 GB	-
JLU/ GAU	Scoring data	Only BBCH10-11	-	Three csv-files with scoring data (see column „Visual scoring finished...”) plus a folder with 470 pictures which contains the most similar/reliable independent scorings from BBCH10 at 21.10. in GG (less than 5% difference between GAU and JLU scoring) which might be used as calibration data set	3.7 GB	Folder RFSB-Phenotyping_training_set.zip (470, subset from 937 pictures)

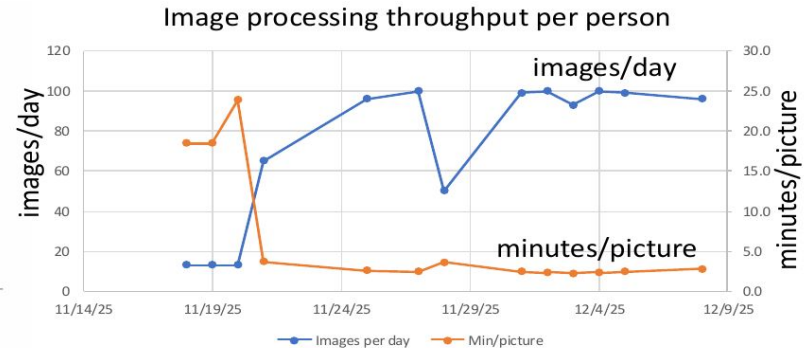
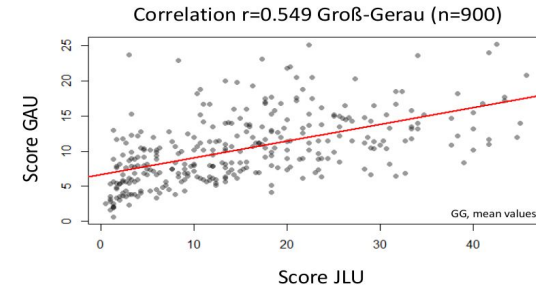
Note: Visual scoring is only finished for part of the data (2747 pictures). For 937 pictures scoring was done twice independently by GAU and JLU. Some scorings were quite different. However, for 470 pictures the (average) scores per picture were similar (less than 5% difference). We suggest to use these pictures as a calibration set. QR codes in the pictures are identical within 3 pictures taken from 3 different positions within the same plot (the same genotype) and QR codes are also identical for the same plots and different sampling time points (1st and 2nd CSFB phenotyping), only file *.jpg names are unique. JLU = Justus Liebig University, GAU = Georg August University

Obstacles To Overcome

- Manual labelling extremely challenging
- Scoring highly subjective
- Only limited labelled data available
- Different conditions (e.g. lighting)



GG, mean values



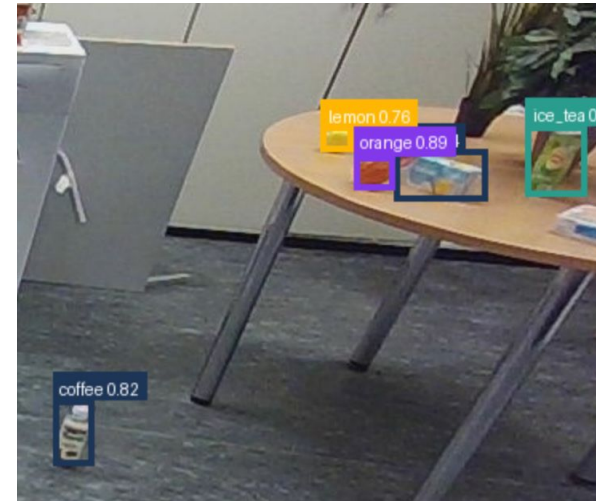
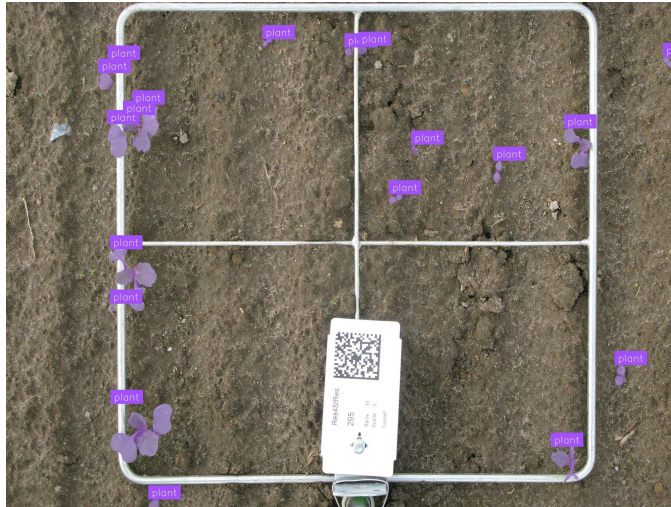
How can we train a ML model that correctly scores plant damage under different conditions and given only little labelled data?

Possible Approaches

- Let us start with **scoring** only (feature computation later)
- Let us focus on **younger plants** for now (BBCH 10-11)
- In the following: Ideas for tackling the problem
- Own ideas are welcome and encouraged!

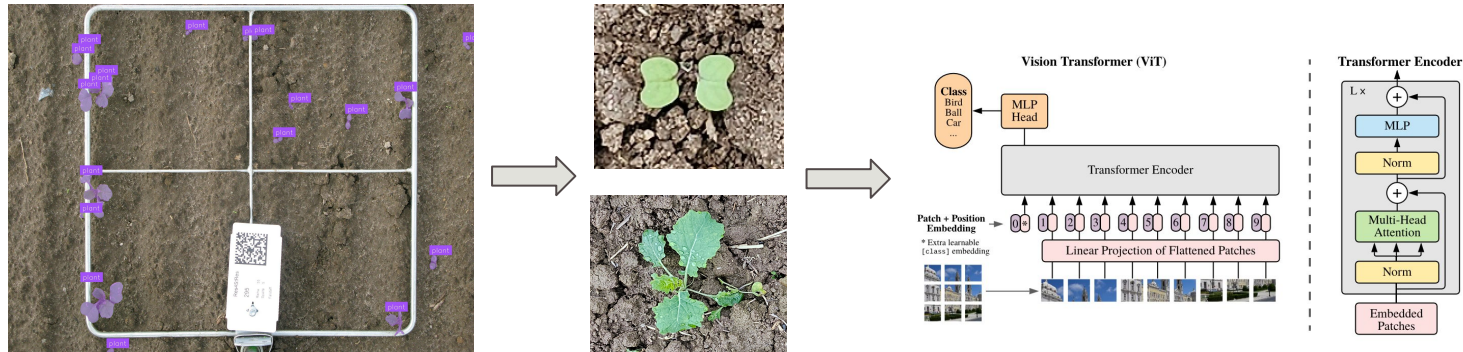
Some useful tools

- SAM 3 for automatic segmentation
- YOLO/RFDETR for object detection



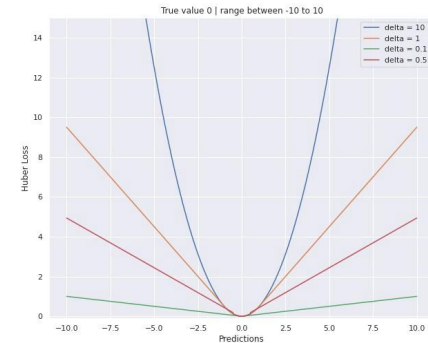
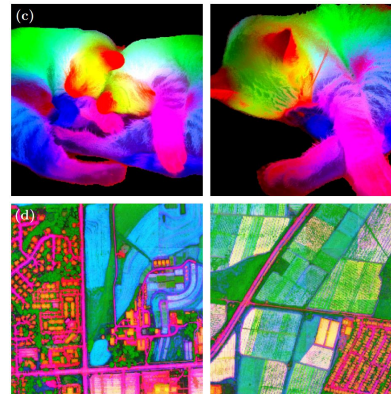
Preprocessing Pipeline: SAM 3

- Use **SAM 3** to segment all plants in an image
- Apply some **filtering** to remove bad results (hardcoded or using a classifier)
- Forward **each plant** through the model
- **Combine** scores of all plants (e.g. weighted average, deep sets, concatenation...)



A baseline approach

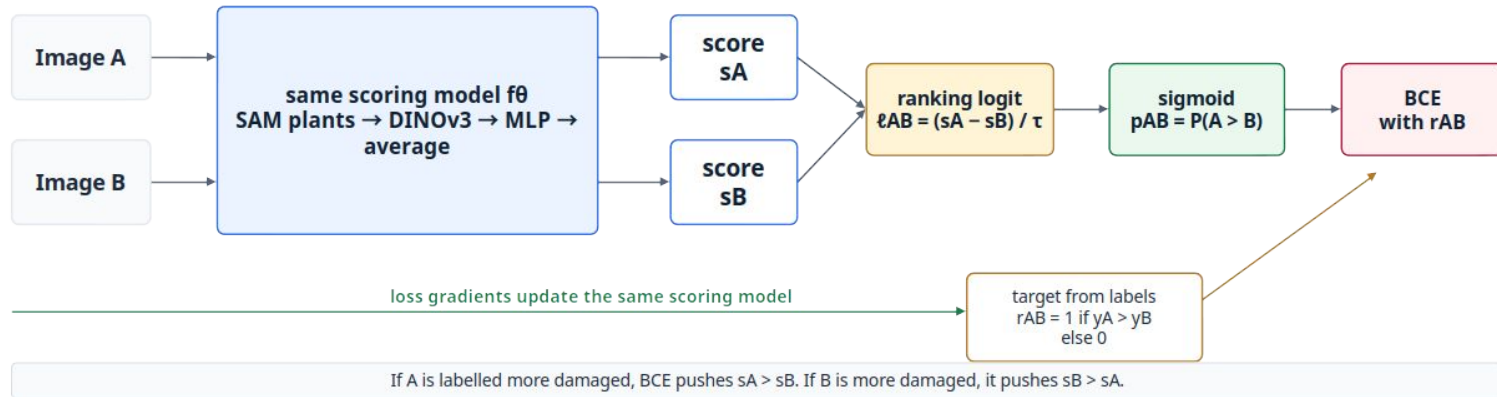
- Standard **supervised** approach
- Use **DINOv3** as frozen backbone and train a **scoring head**
- Training/Validation/Test Data: **470** “high-quality” labelled images
- Further test on different data - from similar data to unsimilar data



Ranking-Based Training

- Idea: Do we really need perfectly labelled scores for training?
- Instead: Train based on comparison
- Example: Image A shows more plant damage than Image B

Pairwise ranking loss: train the same scoring model to order image pairs



BCE = Binary Cross Entropy
 s_A/s_B = scores predicted for images A/B
 τ = temperature tuning parameter
 y_A/y_B = score labels for images A/B

$$p_{AB} = \sigma \left(\frac{s_A - s_B}{\tau} \right),$$

*later we might tackle more sophisticated methods for ranking-based training, e.g. with an own pairwise comparison head

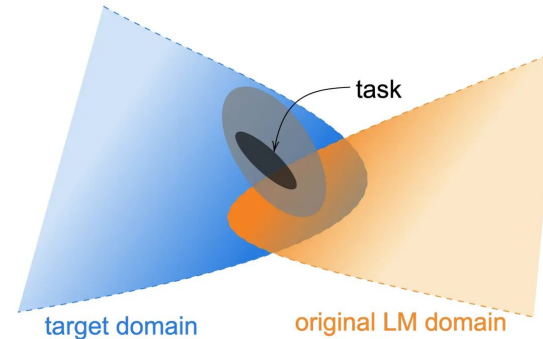
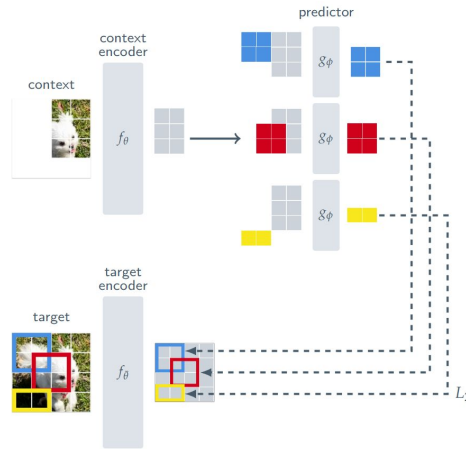
Few-Shot Learning

- Can we create scores (or at least weak labels) using a VLM?
- Prompt a VLM with an example scoring guide and an image and ask it to estimate plant damage
- Depending on how good these scores are, use them as labels for training (or even directly as scores)



Domain-Adaptation

- Use the full dataset for self-supervised pre-training of **DINOv3** or **I-JEPA**
- Again train a scoring head using the **470** labelled images
- Is **Domain-Adaptation** feasible with respect to compute resources?



Your Tasks

1. Choose one of the directions (Ranking-Based, Few-Shot Learning, Domain-Adaptation)
2. **Implement the baseline approach** (Estimated time: 2 weeks)
-> Please add some means of analysis, for instance via visualization / UI / tables
3. Report your results as soon as the baseline approach is finished
4. Meanwhile, you can already start thinking about how to improve the system towards the direction that you choose and implementing it. In the end it would be nice if every team manages to work a bit on each direction.
5. You can also think about how to compute the additional features of interest. For instance, you might try to train a hole / brown pit detector or a plant age classifier